1. Which is the process of change that has transformed life on Earth.
A. Ecosystem
B. Evolution
C. Genetics
D. Taxonomy
Answer: B
2. The development of new tools and techniques has improved the quality of our _____ .
A. living plaints
B. living animal
C. life
D. living organisms
Answer: C
3. What distinguishes horticulture from agriculture?
A. Scale of production
B. Irrigation system
C. Use of technology
D. Geographic location
Answer: C
4. Research in ecology and conservation inform how we can protect this planet's precious _____.
A. environment
B. surrounding
C. forest
D. biodiversity
Answer: D
5. Which level of biological organization involves the interactions between different species living in the
same area?
A. Communities
B. Ecosystem
C. Populations
D. Biosphere
Answer: A
6. The genetic information is encoded in the nucleotide sequences of _____.
A. ATP
B. RNA
C. DNA
D. ADP
Answer: C
7. Among the ASEAN countries, Myanmar is the leading _____ and bean producer.
A. rice
B. pulse
C. maize
D. wheat

Answer: B

8. Hydroponics require _____.
A. purified water
B. mineral nutrient solution
C. inert medium
D. All of these
Answer: D
9. What is the scientific name greengram?
A. Vigna mungo
B. Sesamum indicum
C. Vigna radiata
D. Arachis hypogaea
Answer: C
10. The cultivation of marine organisms in enclosed sections of the open ocean is called
A. offshore mariculture
B. inshore mariculture
C. onshore mariculture
D. integrated fish farming
Answer: B
11. The most requirement for traditional rice cultivation method is _____.
A. fertilizer
B. irrigation planning
C. pest control
D. weed control
Answer: B
12. Which is the growing and marketing of flowers and ornamental plants for floristry.
A. Pomology
B. Olericulture
C. Arboriculture
D. Floriculture
Answer: D
13. Choose the correct match.
A. Tectona grandis – Pyinkado
B. Xylia dolabriformis – Teak
C. Corchorus – Jute
D. Vidna mungo – greengram
Answer: C
14. Structural timbers are obtained mainly from _____.
A. Softwoods
B. Heart of the tree
C. Neither A nor B
D. Both A and B
Answer: D

15. One of the main roles of DNA primase in DNA replication:
A. Synthesize new DNA strands
B. Synthesize short RNA primers
C. Seal nicks in the sugar-phosphate backbone
D. Unwind the double-stranded DNA helix
Answer: B

16. If one strand of DNA is 5'-ATCTGGTGCCAT-3', the opposite strand must be:
A. 3'-TAGACCACGGTA-5'
B. 3'-UAGACCACGGUA-5'
C. 3'-ATCTGGTGCCAT-5'
D. 3'-UTGACCUCGGUA-5'
Answer: A
17. In RNA, which RNA carries anticodons:
A. mRNA
B. tRNA
C. rRNA
D. all of these above
Answer: B
18. The structure of DNA replication fork is:
A. S-shape
B. C-shape
C. Y-shape
D. X-shape
Answer: C
19. The 61 codons that encode _____ specific amino acids:
A. 20
B. 21
C. 22
D. 23
Answer: A
20. Purposes for developing genetically modified plants include:
A. Improved food quality and shelf life
B. Herbicide tolerance
C. Increased nutritional value
D. All of these above
Answer: D
21. Which of the following is the best definition for recombinant DNA?
A. DNA that results from bacterial conjugation
B. DNA that carries genes from different organisms
C. DNA that is produced as a result of crossing over
D. An alternate form of DNA that is the product of a mutation
Answer: B
22. AUG codon is transcribed from the DNA sequence:

A. TAC
B. UAC
C. TAG
D. UAG
Answer: A

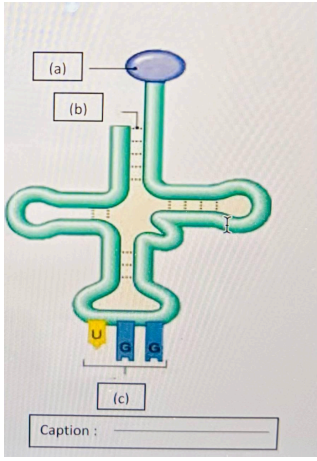
23. How many major steps in PCR:
A. Three
B. Five
C. Four
D. Six
Answer: C
24. In which countries are much produced genetically modified salmon:
A. New Zealand and Australia
B. Norway and Finland
C. France and Switzerland
D. Canada and USA
Answer: D
25. Which type of cell is NOT involved in the process of translocation?
A. Xylem vessels
B. Companion cell
C. Sieve tube
D. Phloem parenchyma cells
Answer: A
26. The ash and sugar content of the fruit increase with the increase in _____.
A. Photosynthesis
B. Transpiration
C. Translocation
D. Respiration
Answer: B
27. Transpiration is the movement or loss of _____.
A. Water from the soil to the roots
B. Water from the roots to the stem
C. Water vapour from the leaves
D. Oxygen from the leaves
Answer: C
28. If water vapour is lost due to excessive transpiration, the amounts of water in its cells wall.
A. Increase
B. Rise
C. Improve
D. Decrease
Answer: D
29. Which of the following statements about symplastic water movement is true?
A. It involves movement through cell walls and intercellular spaces

B. It is typically faster than apoplastic pathway
C. It occurs through the cytoplasm and plasmodesmata
D. It is affected by the Casparian strip
Answer: C
30. What is transpiration?
A. Evaporation of water from mesophyll cells and its diffusion out of stomata
B. Uptake of minerals by root hairs
C. Movement of sap through phloem
D. Conversion of sucrose into starch
Answer: A

SECTION B (10 x 2 = 20 MARKS)
1. Which of the following practices can lead to sustainable food production?
A. Heavy reliance on synthetic fertilizers and pesticides
B. Deforestation for agricultural expansion
C. Implementing crop rotation and organic farming
D. Overgrazing of livestock on rangelands
Answer: C
2. How is spider silk produced using biotechnology?
A. It is extracted from naturally occurring spider webs.
B. The spider silk gene is inserted into goat mammary glands to produce silk proteins in milk.
C. Spider DNA is modified to enhance silk production in insects.
D. Bacteria are genetically engineered to synthesize silk directly.
Answer: B
3. Six processes in genetic engineering are listed.
1 The rice plants are grown.
2 The β-carotene pathway genes are inserted into rice DNA.
3 The genes are isolated from daffodils and bacteria.
4 The rice grains produce β-carotene.
5 The modified rice embryos are cultured.
6 The genes are copied.
Which four processes, in the correct order, show the production of Golden Rice?
A. 3 → 2 → 5 → 4
B. 6 → 3 → 2 → 1
C. 2 → 5 → 1 → 4
D. 3 → 6 → 5 → 1
Answer: A
4. Some events that take place during the translation stage of protein synthesis are listed below:
1. The ribosome moves along the mRNA to the next codon.
2. The completed polypeptide chain is released.
3. A tRNA molecule carrying an amino acid pairs with a specific codon on the mRNA.
4. Peptide bonds form between adjacent amino acids.
5. The ribosome binds to the mRNA strand.
The correct order of these events is:

A. 3 → 5 → 4 → 1 → 2
B. 5 → 4 → 2 → 1 → 3
C. 5 → 3 → 1 → 4 → 2
D. 5 → 3 → 4 → 1 → 2
Answer: D
5. A researcher observes that mRNA is being produced normally, but no protein is being synthesized. Which
of the following is the most likely explanation?
A. Defect in ribosome assembly or function
B. Deletion of an intron
C. Overactive RNA polymerase
D. Mutation in the DNA template strand
Answer: A

6. Give label and caption of the following diagram.



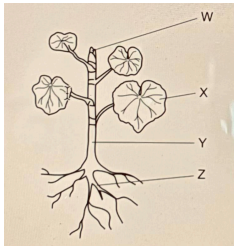
A. (a) – anticodon, (b) – T-loop, (c) – amino acid, caption –rRNAs
B. (a) – amino acid, (b) – D-loop, (c) anticodon, caption – mRNAs
C. (a) – anticodon loop, (b) – amino acid, (c) – codon, caption – RNAs
D. (a) – amino acid, (b) – complementary base paring, (c) – anticodon, caption – tRNAs
Answer: D
7. A mutation alters the promoter region of a gene, reducing the binding efficiency of RNA polymerase.
What is the most likely consequence of this mutation?
A. Increased rate of translation
B. Reduced or no transcription of the gene
C. Enhanced splicing of pre-Mrna
D. Faster addition of poly-A tail
Answer: B
8. A leafy twig is placed in a thistle funnel setup, but the mercury level does NOT rise. Which condition
BEST explains the failure?

A. The leaf surface area is very large
B. Mineral ions are absorbed too quickly
C. Transpiration does not occur because stomata are closed
D. Companion cells are active
Answer: C
9. The table shows the rates of water uptake and transpiration of a plant during a morning.

Time	09:00	10:00	11:00	12:00
Rate of water uptake/ per hour	15	16	16	17
Transpiration /per hour	7	10	12	19

At what time does the plant shows signs of wilting?
A. 09.00
B. 10.00
C. 11.00
D. 12.00
Answer: D

10. The diagram shows a plant.



What is the pathway taken by most of the water absorbed by this plant?
A. X → Y → Z
B. Z → Y → X
C. W → Y → X
D. Z → Y → W

Answer: B